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Thumbwheel 175D

Due to its flat, miniature design, illuminated thumbwheel switch 175D is ideal for use in joystick grips and the keypads of mobile machinery.

The thumbwheel can be mounted in our joystick grips 361G / 362G and 341G / 342G, for the control of operational functions, such as those on wheel loaders

Technical specifications

- Optional LED illumination of various colours
- Ergonomically adaptable by means of various actuator types
- High reliability thanks to shielded, contactless Hall technology
- IP67 electronics protection class
- Deflection: ±10° or ±30° no-detent, ±20° detent
- For wiring and signal form, select between ratiometric analogue, semi-redundant analogue and redundant analogue or digital

Technical drawing

IMAGE 1/3

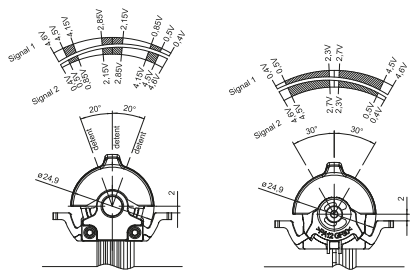


IMAGE 2/3

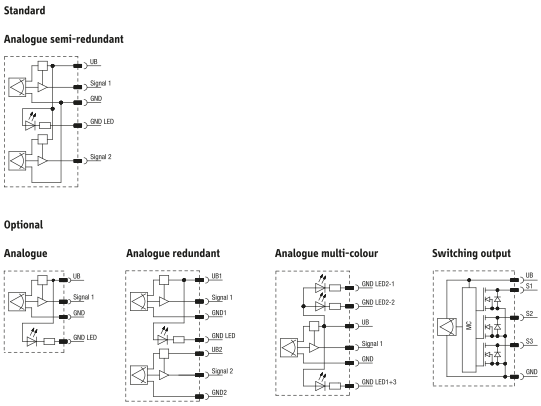
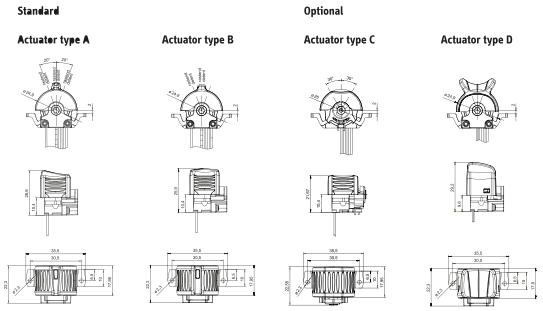


IMAGE 3/3



Product options

IMAGE 1/2

Illumination

Illumination is generally provided by 3 LEDs. An optional multi-colour LED (red-green) is available for the centre position, which maintains its illumination colour until a signal position change takes place. Optionally also available without illumination.



Actuator colors Standard
RAL 2010, signal orange
RAL 9004, signal black

Illumination colours LED
White

Actuator colors Optional
RAL 7043, traffic grey
RAL 3001, signal red

Illumination colours LED Optional
Red
Green
Orange

Red/green for multi-colour LEDs

IMAGE 2/2

ORDERING KEY

175D								Thumbwheel
								Actuation type detent, +/-28°
	R	1	B	1	S	W	0	9	Actuator type B, LED colour white, cap colour black
	R	1	B	1	0	R	0	3	Actuator type B, LED colour white, cap colour orange
	R	1	A	1	S	W	0	6	Actuator type A, LED colour white, cap colour black
	R	1	A	1	0	R	0	9	Actuator type A, LED colour white, cap colour orange
								Actuation type no detent, +/-30°
	T	1	B	1	S	W	3	0	Actuator type B, LED colour white, cap colour black
	T	1	B	1	0	R	0	5	Actuator type B, LED colour white, cap colour orange
	T	1	A	1	S	W	4	1	Actuator type A, LED colour white, cap colour black
	T	1	A	1	0	R	1	6	Actuator type A, LED colour white, cap colour orange

OPTIONS

Optional	
Electronics	Analogue single-channel
	Analogue redundant
	Analogue multi-colour LED
	Switching output
	Various LED illumination colours
Mechanics	Actuation type no detent, +/-10°
Handle	Actuator type C, RAL 9004, signal black
	Actuator type D, RAL 9004, signal black
	Actuator type A / B, RAL 7043, traffic grey
	Actuator type A / B, RAL 3001, signal red

Electrical data

Attribute	175DD...	175DR...	175DT...
Max. switching voltage	24 V DC	-	
Max. switching current	0.1 A	-	
Polarity reversal protection	no		
Output signal min.	-	0.5 V DC	
Output signal max.	-	4.5 V DC	
Output signal	open drain	analogue	
Output signal - centre position/zero position	-	2.5 V	
Output polarity	-	opposite	
Operating voltage min.	4.5 V DC		
Operating voltage max.	5.5 V DC		
Current consumption	per Signal max. 15 mA		
Load resistance min.	-	20000 Ohm	
Short-circuit resistance to GND	yes		
Short-circuit resistance to supply	yes		
Technology	Hall		

Mechanical data

Attribute	175DD...	175DR...	175DT...
Deflection	± 30 °	± 20 °	± 30 °
Actuation type	no detent	detent	no detent
Service life, mechanical (Cycles)	2000000		

Environmental conditions

Attribute	175DD...	175DR...	175DT...
Protection class, electronic	IP67 DIN EN 60529		
Operating temperature min.	-25 °C		
Max. operating temperature	85 °C		
Min. storage temperature	-40 °C		
Max. storage temperature	85 °C		
Shock resistance (Norm)	Frequency 50 Hz - 100Hz - 50 Hz, Acceleration 5 g, Amplitude 15 mm peak		

Installation

Attribute	175DD...	175DR...	175DT...
Installation	from below		

Connection

Attribute	175DD...	175DR...	175DT...
Cable length	0.75 m		